Joins, wreath products and reversal of polytope markings

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Abstract

We apply the common orbit construction to joins, permutation wreath products and reversal of regular polytope markings. Coordinate formulas give the full parabolics and the exact panel sizes, including nontrivial chamber stabilizers. An explicit module automorphism identifies the three pairs of signed permutation constructions in the source. Infinite joins lead to a direct description of the restricted and unrestricted lamplighter geometries. The application records corrections to several unrestricted source statements. The ten affirmative sporadic cases have separate finite certificates [5,6]. Their finite input data and computational proof obligations are recorded there rather than being assumed in the orbit argument.

1 Scope and common input

The original is Piedade and Tranchida's *The geometry of wreath and semi-direct products*, arXiv:2605.04646v2, twenty-two pages [1]. We use the complete companion paper *Orbit intersections and marked semidirect products* [2]. Its orbit construction permits nontrivial Borel subgroups; its faithful regular specialization identifies the full group of type-preserving automorphisms. Its module and cocycle theorems will be used by name. None of those proofs is assumed from the original.

For a coset geometry $\alpha = (A, (A_j)_{j\in S})$ write $A_E = \bigcap_{j\in E} A_j$, with $A_\emptyset = A$. The Borel subgroup is A_S. For a faithful regular hypertope, distinguish the involutions a_j associated with the base chamber and put $A(R) = \langle a_j : j \in R\rangle$. Then $A_E = A(S \setminus E)$. Keeping these two notations separate is essential when a Borel subgroup is nontrivial. All cosets below are left cosets.

The source's preliminary Lemma 2.4 identifies certain generated parabolics with complementary intersections without requiring a trivial Borel subgroup. Its empty generated subgroup is 1, so these equalities cannot hold at the extremes unless the Borel is trivial. A separate generation hypothesis is needed for the converse intersection criterion. For example, in $G = \langle x, y, z\rangle = C_2^3$, the parabolics $\langle y, z\rangle, \langle x, z\rangle$ give the connected geometry $K_{2,2}$ but have intersection $\langle z\rangle \neq 1$. Using instead $\langle x\rangle, \langle y\rangle$, the four generated-support subgroups $1, \langle x\rangle, \langle y\rangle, \langle x, y\rangle$ satisfy every support-intersection identity, although the coset geometry is two disjoint copies of $K_{2,2}$. The complete common paper uses residual generation with the Borel retained, which applies to both faithful and unfaithful models.

2 Joins and their two group models

Let $(\Gamma_\omega)_{\omega\in\Omega}$ be coset incidence systems, with groups H^ω and finite type sets I_ω. Their type sets are regarded as disjoint. The join has their disjoint union of elements, retains incidence within each factor, and declares every pair in distinct factors incident. We allow Ω to be infinite in this section.

Proposition 1. *Both the direct product $\prod_\omega H^\omega$ and the restricted direct product $\bigoplus_\omega H^\omega$ realize the join as a coset incidence system. For a type $i \in I_\lambda$, its maximal parabolic restricts the λ coordinate to H^λ_i and leaves every other coordinate unrestricted, within the chosen product.*

If every factor action is flag-transitive, the full product action is flag-transitive. The restricted product carries one flag to another of the same type whenever they differ in only finitely many factors. Every join of geometries is a geometry; it is residually connected, firm or thin when all factors have the corresponding property.

Proof. Projection to coordinate λ induces a bijection from the cosets of the displayed maximal parabolic to H^λ/H_i^λ. Surjectivity holds for the restricted product as well: a representative can have just its λ coordinate nonidentity. Two cosets with types in the same factor intersect exactly when the projected cosets intersect. For two different factors the coordinate requirements can always be satisfied simultaneously by an element with at most two nonidentity coordinates. These statements identify both incidence systems with the join, without any finiteness assumption on Ω.

A flag is the union of its component flags. Extend each component to a chamber; their union is a chamber of the join. Conversely a maximal flag must be maximal in each component. Choose, in each coordinate, an element carrying one component flag to the other. Their tuple lies in the full product. If only finitely many component flags differ, take the identity in every other coordinate to obtain a restricted-product element. These choices use the usual axiom of choice for arbitrary families.

A residue is the join of the corresponding factor residues, after discarding those with no missing types. If at least two nonempty factors remain, the incidence graph is connected because any two vertices in one factor are connected through a vertex of another factor. If exactly one factor remains and the residue has rank at least two, connectivity is the residual connectivity of that factor. Finally a panel misses one type in one factor, so its number of completions is exactly that of the corresponding factor panel. This proves the last assertions. □

For two finite-rank factors the group is $A \times B$, and direct intersection gives

$$G_J = A_{J\cap S} \times B_{J\cap I}.$$

Its geometry is finite exactly when both factors are finite. These observations recover source Lemma 3.2 and Theorems 3.3, 3.6 and 3.7. The source's assertion that the restricted product is *never* flag-transitive for infinite Ω needs a qualification. It fails to be flag-transitive if infinitely many factors permit distinct flags of the same type: choose such flags in every one of those factors, and no finitely supported element can carry their unions to each other. An infinite family of one-chamber factors is a counterexample to the unqualified assertion. In the firm examples used below every nonempty factor does permit different chambers, so the stated obstruction applies.

3 Wreath products and exact panel sizes

Let $\alpha = (A, (A_j)_{j\in S})$ and $\beta = (B, (B_i)_{i\in I})$ be flag-transitive geometries of finite, nonzero rank. Let Ω be a finite nonempty B-set. For every orbit $O \subseteq \Omega$ and every $j \in S$, choose a point $\lambda_{O,j} \in O$ satisfying

$$B_H\lambda_{O,j} \cap B_K\lambda_{O,j} = B_{H\cup K}\lambda_{O,j} \qquad (H, K \subseteq I). \tag{1}$$

The choices may depend on j. Let $D = A^\Omega$, with $b(v)_\omega = v_{b^{-1}\omega}$, so $D \rtimes B = A \wr_\Omega B$. The type set of the join α^Ω is $S \times \Omega$. Its orbit types are $L_{O,j} = \{j\} \times O$.

Proposition 2. *The coordinate action is admissible for the common orbit construction exactly when the selected representatives satisfy* (1)*. The wreath geometry has the following maximal parabolics. For $i \in I$ put*

$$E_i(\omega) = \{j \in S : \omega \notin B_i\lambda_{O(\omega),j}\}.$$

Then

$$G_i = \left(\prod_{\omega\in\Omega} A_{E_i(\omega)}\right) \rtimes B_i, \qquad i \in I, \tag{2}$$

$$G_{L_{O,j}} = \left(\prod_{\omega\in O} A_j \times \prod_{\omega\notin O} A\right) \rtimes B. \tag{3}$$

It is a flag-transitive geometry, residually connected and firm if the inputs are so. Its old panel sizes are $[B_{I\setminus\{i\}} : B_I]$*, and its new panel size at* $L_{O,j}$ *is*

$$[A_{S\setminus\{j\}} : A_S]^{|B_I\lambda_{O,j}|}. \tag{4}$$

In particular, for thin inputs it is thin exactly when B_I *fixes all the selected representatives.*

Proof. The action carries a coordinate parabolic indexed by (j, ω) to that indexed by $(j, b\omega)$. Its orbit is $L_{O,j}$, with the same orbit intersection identity as the action on O. This proves admissibility and its converse with the indexed action understood.

The common theorem takes the intersection of coordinate parabolics outside the union of the orbits $B_i(j, \lambda_{O,j})$. At coordinate ω the retained restrictions are exactly the $j \in E_i(\omega)$, which proves (2). Intersecting every coordinate parabolic in $L_{O,j}$ gives (3). The common theorem and Proposition 1 give flag transitivity, residual connectivity and firmness.

The Borel subgroup of the product geometry is A_S^Ω. Deleting the new type $L_{O,j}$ relaxes the restriction from A_S to $A_{S\setminus\{j\}}$ precisely at the coordinates $B_I\lambda_{O,j}$. Taking the index over A_S^Ω gives (4). For thin inputs its base is two, so the index is two if and only if that nonempty orbit has one point. The old panels already have size two. This proves the equivalence. □

If representatives are chosen coherently, so $\lambda_{O,j} = \lambda_O$ for every j, the coordinate factor in (2) is A on $\bigcup_O B_i\lambda_O$ and A_S outside it. The printed source Proposition 4.3 reverses this support and omits the Borel factors. The formula above follows directly from its defining intersections and also covers independent choices for different types.

Source Theorem 4.4 asserts thinness without the necessary orbit condition. Take $\alpha = (C_2, 1)$, $\beta = (S_3, C_3)$ of rank one and let S_3 act on three coordinates. Both geometries have two elements and are thin. The orbit identity is automatic for rank one. But $|C_3\lambda| = 3$, so the new panel has size $2^3 = 8$. More explicitly, the two maximal parabolics are $C_2^3 \rtimes C_3$ and S_3 in a group of order 48. Their coset geometry is the complete bipartite graph $K_{2,8}$: the product of the parabolics is the whole group. It is firm and connected, but not thin. Passing from S_3 to S_3/C_3 does not repair this construction: the original action on three coordinates does not factor through that quotient.

For faithful regular inputs the Borel groups are trivial. Every permitted wreath construction is then a regular hypertope, with full type-preserving automorphism group $A \wr_\Omega B$. This is the precise regular case used in the source applications.

4 The six signed permutation families

Let $r \geq 2$ and represent $\mathbb{F}_2^r \rtimes S_r$ on $2r$ points in r blocks of size two. A vector e_j flips block j and the adjacent transposition $s_i = (i\ i+1)$ interchanges two adjacent blocks without internal flips. Put $\lambda = \sum e_j$. The complete common module theorem applies to each seed $\lambda, e_1, \lambda + e_1$; the first mark is its block-flip vector and the remaining marks are $s_1, \ldots, s_{r-1}$.

Corollary 3. *All six lists obtained by using one of these three seeds and either the marks* $(0, s_i)$ *or the marks* $(\lambda + e_i + e_{i+1}, s_i)$ *are string C-groups. They give the diagrams (1), (2) of Table 9 and (13)–(16) of Table 6 in the bound Fernandes–Piedade catalogue [3]. Their abstract groups, in the order of the three seeds, are* $C_2 \times S_r$*,* $C_2 \wr S_r$*, and* $C_2 \wr S_r$ *for even* r *or* $\mathbb{F}_2^{r,\mathrm{even}} \rtimes S_r$ *for odd* r*.*

Proof. The ordinary markings satisfy every subgroup intersection by the common module theorem. Its change of coordinates fixes the first mark and sends each $(0, s_i)$ to $(\lambda + e_i + e_{i+1}, s_i)$. That latter vector means exactly the vertical i-edges in every block except i and $i+1$ in the source diagrams. Thus this is an identification of the actual marked permutations, not merely of their abstract groups. The three vector spans and their sole odd-rank relation are calculated in the common proposition on signed markings. Transport through its automorphism proves the assertion. □

The auxiliary classification in [3] has range $r = n/2 \geq 7$; the algebraic construction just proved works for every $r \geq 2$. Source Proposition 5.1 covers five of these markings in both parities and the sixth only for even r. The preceding paragraph of that source leaves odd-rank graph (16) unverified. Corollary 3 proves its intersection property too. This is a solution of that explicitly recorded gap; it is not a historical-priority assertion. It does not prove the external classification of all possible high-rank permutation graphs.

5 Reversal of a regular polytope

Let α be a self-dual regular polytope of rank $n \geq 3$, with distinguished involutions $a_0, \dots, a_{n-1}$ and group A. A duality reverses types. Composing a duality with an automorphism of α makes it preserve the base chamber as a set. Its square is type-preserving and fixes that chamber, so is the identity by regularity. Conjugation therefore gives an involution $\tau \in \mathrm{Aut}(A)$ satisfying

$$\tau(a_i) = a_{n-1-i}. \tag{5}$$

It is nonidentity because the distinguished involutions are distinct.

Proposition 4. *Choose one representative from each orbit of the reversal $i \mapsto n-1-i$. Together with the outer factor t of order two, the selected a_i are the distinguished involutions of a regular hypertope for $A \rtimes \langle t \rangle$, where t acts by τ. Its rank is $\lceil n/2 \rceil + 1$. For selected i, j their mutual label is* $\mathrm{ord}(a_i a_j)$. *Their labels with t are*

$$\mathrm{ord}(a_i t) = \begin{cases} 2, & i = n-1-i, \\ 2\,\mathrm{ord}(a_i a_{n-1-i}), & i \neq n-1-i. \end{cases}$$

For $n = 3, 4, 5$ representatives can always be chosen so that the marking is a string C-group. For $n \geq 6$ its diagram is non-linear for every choice of representatives. For $n = 4, 5$, a non-linear choice exists when $\mathrm{ord}(a_0 a_1) > 2$.

Proof. Use the rank-one geometry $(\langle t \rangle, 1)$ in the faithful regular orbit construction. For a rank-one group, the only parabolics are the whole group and the identity, and their orbit intersection identity holds at every point. There are $\lceil n/2 \rceil$ reversal orbits. The common theorem gives the asserted geometry, generation and full automorphism group. Its exact doubled-label lemma gives the displayed order for a nonfixed index. At a fixed index t commutes with a_i; their product is a nonidentity involution since it projects to t.

For $n = 3$ choose $0, 1$. The mark a_1 commutes with t, giving the string order a_1, a_0, t. For $n = 4$ choose $0, 2$. The old marks commute, giving the order a_0, t, a_2. The source's printed choice $0, 3$ selects the same reversal orbit twice; $0, 2$ is the required correction. For $n = 5$ choose $0, 2, 3$, with string order a_0, t, a_3, a_2. Every nonconsecutive pair commutes, by the displayed orders and the original string relations. The ordinary correspondence between string C-groups and regular polytopes gives the polytope claims, including possible labels two.

Every nonfixed reversal orbit contributes a neighbour of t in the diagram. Indeed a_i and a_{n-1-i} are distinct involutions, so their product has order at least two and its doubled label is at least four. There are $\lfloor n/2 \rfloor$ such orbits. For $n \geq 6$ the degree of t is therefore at least three. For $n = 4, 5$ select $0, 1$ and also the fixed middle index when present. If $\mathrm{ord}(a_0 a_1) > 2$, the three nodes a_0, a_1, t form a triangle. In rank five the full diagram is a triangle with an additional edge to the middle mark when that label exceeds two; it is not the square described in the source. □

The unconditional non-linear assertion of source Theorem 5.2(b) is false in rank four. To specify an admissible all-digon polytope of rank n, take two faces at each proper rank and one least and greatest face. Every face of lower rank is below every face of higher rank. Intervals of rank difference two have two intermediate faces. Chambers are independent binary choices at the n proper ranks. Changing one

choice at a time proves strong flag connectivity, including while retaining any fixed common faces. Thus this poset is an abstract regular polytope. Its full automorphism group independently swaps the two faces at each rank, so is C_2^n, with the standard basis as marks. Rank reversal is a duality fixing the chamber of zero choices.

Now take $n = 4$. In $A \rtimes \langle t \rangle$, all involutions in A commute. An involution outside A has form vt with $\tau(v) = v$; any two such involutions commute as well. Every triple of involutions therefore contains a commuting pair. In particular no rank-three hypertope with triangular diagram has this group. The common construction and the polytope conclusion remain valid; the lost claim is the unrestricted non-linear choice. Indeed every regular hypertope has distinguished involutions, and a commuting pair gives a rank-two residue of gonality two, hence no edge between those types. A non-linear diagram on three nodes would require all three edges.

Lemma 5. *Suppose τ is not inner. The natural map*

$$A \rtimes \langle t \rangle \longrightarrow \mathrm{Aut}(A), \qquad (a, t^\epsilon) \longmapsto \mathrm{Inn}(a)\tau^\epsilon$$

has kernel $Z(A) \times 1$. It is injective exactly when A is centerless.

Proof. Conjugation intertwines the action of τ, so the displayed map is a homomorphism. A kernel element with $\epsilon = 1$ would express τ as an inner automorphism. With $\epsilon = 0$ its image is the identity exactly when a is central. This proves both statements. □

Source Theorem 5.2(d) omits the center. For the rank-three all-digon polytope the reversal is outer, the semidirect group has order 16, and $|\mathrm{Aut}(C_2^3)| = |\mathrm{GL}_3(2)| = 7 \cdot 6 \cdot 4 = 168$. Lagrange's theorem rules out even an abstract embedding. For sporadic simple groups the centerless hypothesis does hold.

Nor does $\mathrm{Out}(A)$ automatically act on A: it is a quotient of $\mathrm{Aut}(A)$, and a lift as a group action is additional data. For $A = D_{16}$ the quotient map $\mathrm{Aut}(A) \to \mathrm{Out}(A)$ has no section. Here is a direct proof, also obtained independently in [4]. Write $A = \langle r, s \mid r^8 = s^2 = 1, srs = r^{-1} \rangle$. Every automorphism has form $f_{a,b} : r \mapsto r^a, s \mapsto r^b s$, where $a \in \{1, 3, 5, 7\}$ and $b \in \mathbb{Z}/8$. This follows because $\langle r \rangle$ is the unique cyclic subgroup of order eight; all these substitutions do give automorphisms. Composition is $f_{a,b} f_{c,d} = f_{ac,b+ad}$. The inner subgroup has $a = 1, 7$ and even b, so the outer quotient is C_2^2, generated by the classes $X = [f_{1,1}]$ and $Y = [f_{3,0}]$. An involutory lift of X is $f_{7,c}$ with c odd. An involutory lift of Y is either $f_{3,b}$ with b even, or $f_{5,b}$ with $b = 0, 4$. Commutation with the first kind requires $c - 3b = 0$ modulo four, impossible by parity; with the second kind it requires $4c - 6b = 0$ modulo eight, also impossible. A section would provide commuting involutory lifts of X, Y, a contradiction. Throughout this application we specify actual automorphisms such as (5).

6 Two markings of the seven-generator Coxeter example

In source Example 3.9 the Coxeter diagram is a three-armed tree with edges $01, 12, 03, 34, 05, 56$, all labelled three. All other pairs commute. Let b_7 interchange arms $(1, 2)$ and $(3, 4)$ and let b_8 interchange arms $(3, 4)$ and $(5, 6)$. They generate S_3, acting by Coxeter diagram automorphisms. Its type orbits are $\{0\}$, $\{1, 3, 5\}$ and $\{2, 4, 6\}$.

For the natural action of S_3 on three points, its two rank-one parabolics are the subgroups generated by the two adjacent transpositions. At any selected point, their orbits intersect only in that point: at an endpoint one orbit is a singleton, and at the middle point the two two-element orbits have only the middle point in common. This proves the full orbit identity, since the other parabolics are 1 and S_3. Thus any choice of one representative from each orbit is admissible.

For representatives $0, 1, 2$, the old maximal parabolics are

$$G_7 = A_{\{3,4,5,6\}} \rtimes \langle b_8 \rangle, \qquad G_8 = A_{\{5,6\}} \rtimes \langle b_7 \rangle,$$

and the three orbit parabolics are $A_L \rtimes S_3$. The resulting five marks have the following noncommuting pairs:

$$a_0a_1 : 3, \quad a_1a_2 : 3, \quad a_1b_7 : 4, \quad a_2b_7 : 4, \quad b_7b_8 : 3.$$

Every omitted pair commutes. The doubled labels follow because a_1 commutes with a_3 and a_2 with a_4 in the Coxeter group. For representatives $0, 1, 6$, the five marks instead form the path

$$a_0 \overset{3}{-} a_1 \overset{4}{-} b_7 \overset{3}{-} b_8 \overset{4}{-} a_6.$$

The common regular theorem proves the complete intersection and geometry properties in both cases, rather than just the pair relations. In the second case the string correspondence supplies a regular polytope.

The parabolic change noted in the source is also exact. Replacing representative 2 by 4 changes G_7 to $A_{\{2,3,5\}} \rtimes \langle b_8 \rangle$. The first G_7 is $S_4 \times C_2$, since the support $\{0, 1, 2\}$ is an A_3 path fixed by b_8. The changed G_7 is $S_3 \times D_8$: its support $\{0, 1, 4, 6\}$ has an A_2 edge and two isolated vertices, and b_8 swaps the latter. They are nonisomorphic, as their abelianizations have orders four and eight. The Coxeter parabolic embedding theorem is the retained classical input in these identifications.

7 The tetrahedron and the cube

For the tetrahedron take $A = S_4$, $a_0 = (1\ 2)$, $a_1 = (2\ 3)$, $a_2 = (3\ 4)$, and let t act by conjugation by $(1\ 4)(2\ 3)$. The representatives $0, 1$ give marks a_1, a_0, t of string type $\{3, 4\}$ (the dual ordering has type $\{4, 3\}$). The resulting group has order 48. Its maximal parabolics have orders $4, 8, 6$, and therefore its three element counts are $12, 6, 8$ in the original orbit/old-type order.

Here is an incidence identification independent of a group-order name. The eight cube vertices are the four tetrahedron vertices and its four faces. An edge is an incident pair (v, F) with v a vertex of the face F; a cube face is a tetrahedron edge e. Its boundary comprises the two endpoints of e and the two faces containing e, with all four corresponding incident pairs. This is a quadrilateral. Put the tetrahedron at the four cube vertices whose sign-coordinate product is $+1$. Associate a tetrahedron face to the antipode of its omitted vertex. A tetrahedron vertex lies in that face exactly when the two associated cube vertices differ in one coordinate. The six tetrahedron edges yield the six coordinate faces of the cube by the preceding rule. These bijections preserve every incidence, proving the cube identification in source Example 3.18. Exchanging point and face types gives its dual polytope ordering.

8 The two lamplighter geometries

Let D be either $D_{\text{fin}} = \bigoplus_{\mathbb{Z}} C_2$ or $D_{\text{all}} = \prod_{\mathbb{Z}} C_2$, written as binary lamp configurations. Let t shift coordinates and let e_n toggle lamp n. In $G = D \rtimes \langle t \rangle$ take maximal parabolics $P = \langle t \rangle$ and $Q = \langle e_0 \rangle$.

Proposition 6. *The coset geometry $(G; (P, Q))$ is the subdivision of the graph whose vertices are configurations in D and whose edges join configurations differing at exactly one lamp. It is flag-transitive. Its connected components are indexed by D/D_{fin}. In particular it is connected for restricted configurations and disconnected for unrestricted configurations. Every edge-type element is incident with two configuration elements; every configuration is incident with countably infinitely many edge-type elements.*

Proof. Write a group element as (v, n), with product $(v, n)(w, m) = (v + t^n w, n + m)$. Its P-coset is determined by v, giving the configuration vertices. Its Q-coset consists of (v, n) and $(v + e_n, n)$; it records a configuration with its nth lamp left unspecified. Incidence with a P-coset means resolving that lamp in one of its two ways. This proves the description and all valencies directly.

Any incident pair of cosets has a common representative. Left translation takes the standard incident pair to that pair, proving flag transitivity in rank two. A path changes finitely many lamps.

Conversely, if two configurations differ on a finite set, toggle those lamps in turn to obtain a path. Components are consequently precisely the cosets of D_{fin} in D. Each unspecified-lamp element lies in the component of its two resolutions. Finally unrestricted configurations can differ at infinitely many lamps, so there is more than one component in that case. □

For completeness the restricted group has presentation

$$\langle a, t \mid a^2 = 1,\ (at^n at^{-n})^2 = 1\ (n \geq 1)\rangle.$$

Indeed the conjugates $a_n = t^n a t^{-n}$ are involutions. The displayed relations and their conjugates make every two of them commute, so every word has the form $\prod_{n\in F} a_n\, t^k$ with F finite. The concrete semidirect product satisfies the presentation, and in it that normal form is the identity only for $F = \varnothing, k = 0$. The mutually inverse maps therefore give the claimed presentation. The source's displayed presentation on page 20 omits the square on $at^n at^{-n}$. Read literally, its $n = 1$ relator makes a commute with t, and the presented group is $C_2 \times \mathbb{Z}$. The squared relators above correct that auxiliary presentation. We do not need or reprove the separate classical theorem that the restricted lamplighter group is not finitely presentable.

This description recovers source Section 5.3 and pinpoints the failure of the finite-type residual-connectivity theorem for unrestricted infinite joins. The group generated by P and Q is only $D_{\mathrm{fin}} \rtimes \mathbb{Z}$, a proper subgroup of $D_{\mathrm{all}} \rtimes \mathbb{Z}$. The missing hypothesis cannot be removed by passing to the full direct product.

9 Attribution and certificates

The constructions, examples and requested results are those of [1]; the six permutation diagrams come from [3]. The join description and the definition of generalized twisting are retained constructions. The common proof [2] supplies the complete orbit argument, the module intersection proof and its explicit change of coordinates. The present paper supplies their coordinate and geometric applications. The correction note [4] gives elementary countermodels for the unqualified thinness, non-linear and automorphism-embedding claims. These are corrections to assertions, not newly solved open conjectures.

The generic cocycle automorphism and transport of subgroup intersections have scoped Lean proofs. They do not formalize the complete geometric arguments, the source diagram identifications or the classification of sporadic groups. Finite validation of signed markings is supplementary; the arbitrary-rank assertion is the written module theorem. The ten affirmative sporadic cases in source Theorem 5.3 are supplied by [5,6], including the Harada–Norton case absent from the source proof. Each certificate establishes both a rank-three regular polytope and a non-linear rank-three regular hypertope. These certificates retain their named group data and, where used, the specified maximal-subgroup classification; those inputs are not consequences of the orbit theorem. The source exceptions concerning Fi_{22} and Fi'_{24} are outside this paper's recovery contract. No length factor or claim of complete formalization follows from the number of cases.

References

[1] C. A. Piedade and P. Tranchida, *The geometry of wreath and semi-direct products*, arXiv:2605.04646v2, 2026.

[2] Human Compression Project, *Orbit intersections and marked semidirect products*, companion proof edition.

[3] M. E. Fernandes and C. A. Piedade, *Regular polytopes of rank $n/2$ for transitive groups of degree n*, arXiv:2407.16003v2, 2025; J. Algebra 689 (2026), 256–296.

[4] Human Compression Project, *Borel subgroups and duality in coset twisting*, independently prepared five-page proof supplement. Its correction proofs used here are included in full above.

[5] Human Compression Project, *Finite witnesses for nine sporadic extensions*, companion certificate packet.

[6] Human Compression Project, *A finite witness for the Harada–Norton extension*, companion certificate packet.